

LUCAS CONGRUENCES FOR APÉRY-LIKE SEQUENCES AND THEIR SECOND SOLUTIONS

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ABSTRACT. Every statement below is proved in Lean 4 and checked by its kernel. Each is followed by the name of the declaration that proves it.

Throughout, p is prime. For $q \in \mathbb{Q}$ write v_p for the p -adic valuation, and $x \equiv y \pmod{p}$ for $x, y \in \mathbb{Q}$ to mean $v_p(x - y) \geq 1$. PInt, PDvd, PCong

1. FIRST SOLUTIONS

Definition 1.1. $A(n, k) = \binom{n}{k}^2 \binom{n+k}{k}^2$ and $a_n = \sum_{k=0}^n A(n, k)$. A, apery

Definition 1.2. $T(n, k, l) = \binom{n+k}{n} \binom{n}{k}^2 \binom{n+l}{n} \binom{n}{l}^2 \binom{n+k+l}{n}^2$ and $Q_n = \sum_{k=0}^n \sum_{l=0}^n T(n, k, l)$. T, Q

Lemma 1.3. For $r, s < p$,

$$\binom{ap+r}{bp+s} \equiv \binom{a}{b} \binom{r}{s} \pmod{p}, \quad \binom{r+s}{s} \equiv 0 \pmod{p} \text{ if } r+s \geq p, \ s \leq r.$$

choose_digits, choose_digits_zero, choose_carry_zero

Lemma 1.4. For $r, s < p$, $A(ap+r, bp+s) \equiv \binom{a}{b}^2 \binom{a+b}{b}^2 A(r, s) \pmod{p}$, and $T(ap+r, bp+s, cp+t) \equiv T(a, b, c) T(r, s, t) \pmod{p}$ for $r, s, t < p$. A_digits, T_digits

Theorem 1.5. For $r < p$ and all $a \geq 0$,

$$a_{ap+r} \equiv a_a a_r \pmod{p}, \quad Q_{ap+r} \equiv Q_a Q_r \pmod{p}.$$

Consequently $a_n \equiv \prod_i a_{n_i}$ and $Q_n \equiv \prod_i Q_{n_i} \pmod{p}$, where $n = \sum_i n_i p^i$ is the base- p expansion. *apery_lucas, Q_lucas, apery_lucas_digits, Q_lucas_digits*

2. DESCENT FOR TRUNCATED TWISTED ZETA VALUES

Definition 2.1. For $\chi : \mathbb{N} \rightarrow \mathbb{Z}$ and $r, y \in \mathbb{N}$, $K_\chi^{(r)}(y) = \sum_{m=0}^y \frac{\chi(m)}{m^r}$, with the term $m=0$ read as 0. K

Lemma 2.2. Let χ be completely multiplicative and $r \geq 1$. Then for all y ,

$$p^r K_\chi^{(r)}(y) \equiv \chi(p) K_\chi^{(r)}(\lfloor y/p \rfloor) \pmod{p}.$$

K_descent

Lemma 2.3. Let $\ell_1, \dots, \ell_t$ be letters of weights $r_1, \dots, r_t$ and characters $\chi_1, \dots, \chi_t$. Then $p^{r_1+\dots+r_t} \prod_i K_{\chi_i}^{(r_i)}(y_i) \equiv \left(\prod_i \chi_i(p) \right) \prod_i K_{\chi_i}^{(r_i)}(\lfloor y_i/p \rfloor) \pmod{p}$. *mon_descent*

3. SECOND SOLUTIONS

Definition 3.1. A *letter* is a triple $\ell = (r, \chi, x)$ with $r \geq 1$, $\chi : \mathbb{N} \rightarrow \mathbb{Z}$ completely multiplicative, and $x : \mathbb{N}^2 \rightarrow \mathbb{N}$; its value is $\ell(n, k) = K_\chi^{(r)}(x(n, k))$. A *monomial* is a finite list of letters, evaluated by multiplication; its degree is $\sum r_i$ and its character value at p is $\prod \chi_i(p)$. Letter, monVal, monDeg, monChi

Definition 3.2. Given $S : \mathbb{N}^2 \rightarrow \mathbb{Z}$, a finite index set J , coefficients $c : J \rightarrow \mathbb{Q}$ and monomials $\text{mon} : J \rightarrow \text{List Letter}$, put $w(n, k) = \sum_{j \in J} c_j \text{mon}_j(n, k)$ and

$$A(n) = \sum_{k=0}^n S(n, k), \quad B(n) = \sum_{k=0}^n S(n, k) w(n, k).$$

W, Arow, Brow

Theorem 3.3. Fix $w \in \mathbb{N}$ and $\chi_p \in \mathbb{Z}$. Assume

- (H1) $S(ap + r, bp + s) \equiv S(a, b)S(r, s) \pmod{p}$ for $r, s < p$;
- (H2) $S(n, k) = 0$ for $k > n$;
- (H3) for $r, s < p$, $b \leq a$ and $S(r, s) \not\equiv 0 \pmod{p}$: $\lfloor x(ap + r, bp + s)/p \rfloor = x(a, b)$ for every letter argument x occurring in mon ;
- (H4) $x(n, k) \leq n$ for every such x and all $k \leq n$;
- (H4c) every c_j is p -integral;
- (Hw) every mon_j has degree w ;
- (H5) every mon_j has character value χ_p at p .

Then for $a, r < p$,

$$p^w B(ap + r) \equiv \chi_p B(a) A(r) \pmod{p}.$$

theorem_LB

4. TWO INSTANCES

Definition 4.1. $H_n^{(r)} = \sum_{j=1}^n j^{-r}$, and

$$b_n = \sum_{k=0}^n A(n, k) (2H_n^{(3)} - H_k^{(3)}), \quad f_n = \sum_{k=0}^n \binom{n}{k}^3,$$

$$\beta_n = \sum_{k=0}^n \binom{n}{k}^3 \left(\frac{1}{4} H_k^{(2)} + \frac{3}{4} (H_k^{(1)})^2 - \frac{3}{4} H_k^{(1)} H_{n-k}^{(1)} \right).$$

Harm, bMin, franel, bFranel

Theorem 4.2. For $a, r < p$,

$$p^3 b_{ap+r} \equiv b_a a_r \pmod{p}, \quad \text{and, for } p \neq 2, \quad p^2 \beta_{ap+r} \equiv \beta_a f_r \pmod{p}.$$

bMin_lucas, bFranel_lucas

5. APÉRY'S COMPANION

Definition 5.1. $P(n) = 34n^3 + 51n^2 + 27n + 5$, and $b^A : \mathbb{N} \rightarrow \mathbb{Q}$ is defined by

$$b_0^A = 0, \quad b_1^A = 6, \quad (n+2)^3 b_{n+2}^A = P(n+1) b_{n+1}^A - (n+1)^3 b_n^A.$$

Ppol, bApery

Lemma 5.2. $(m+2)^3 b_{m+2} - P(m+1) b_{m+1} + (m+1)^3 b_m = 0$ for all m .

bMin_rec

Theorem 5.3. $b_n = b_n^A$ for all n ; that is,

$$\sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2 (2H_n^{(3)} - H_k^{(3)}) = b_n^A.$$

bMin_eq_bApery, apery_b_harmonic_closed_form

Theorem 5.4. For $a, r < p$, $p^3 b_{ap+r}^A \equiv b_a^A a_r \pmod{p}$.

bApery_lucas

6. A DENOMINATOR DEFECT

Definition 6.1. $d_n = \text{lcm}(1, \dots, n)$, and $P_2 = 1190161/384$.

dlcm, BZ_P2

Theorem 6.2. $d_2^5 P_2 = 1190161/12 \notin \mathbb{Z}$, while $12 d_2^5 P_2 = 1190161 \in \mathbb{Z}$; and if $c \in \mathbb{N}$ satisfies $c d_2^5 P_2 \in \mathbb{Z}$ then $12 \mid c$. *BZ_P2_not_integral, BZ_P2_factor_twelve, BZ_twelve_minimal*

7. THE DEVELOPMENT

Lean 4 (v4.33.0-rc1) with Mathlib cd580e54; ten modules, 2107 lines; no `sorry` and no `native_decide`. Every declaration cited above satisfies

`#print axioms` $\longrightarrow$ `[propext, Classical.choice, Quot.sound]`.

Build with `lake exe cache get && lake build`.

REFERENCES

- [1] The mathlib Community, *The Lean mathematical library*, CPP 2020, 367–381. [arXiv:1910.09336](#).
- [2] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, CADE 28, LNCS 12699, 625–635.